Work
(6.4)

Suppose a force acts on an object which is moving in a line. The **work** is the energy transfered to the object by the force. If the force is **constant**, then the work equals

$$\underbrace{\text{force}}_{\text{can be positive or negative depending on direction}} \cdot \underbrace{\text{displacement}}_{\text{change in position of the object}}$$

The units of work are

SI: joules (J), i.e. $kg \cdot m^2/s^2$ or $N \cdot m$ (newtons)

US: foot-pound (ft·lb)

An example of a constant force on an object is Earth's gravity. The weight of an object is the force of Earth's gravity on that object.

- In SI, usually the mass (kg) of an object is given, and we multiply by Earth's gravitational constant $g \approx 9.8\ m/s^2$ to get the weight in Newtons (N)
- In US, usually the weight (lb) of an object is given, so we don't multiply by g.

Example: I hold a 2 lb ball 3 ft above the ground. I throw the ball upward with a constant 7 lb of force, and it leaves my hand at 5 ft above the ground. The ball flies upward until it reaches 10 ft, then it falls to the ground.

Let y be the vertical direction, with "up" being positive.

Between $y=3$ and $y=5$, I act on the ball with constant +7 lb of force, and ball moves +2 ft. So the work done is $(7\text{ lb})(2\text{ ft}) = 14\text{ ft}\cdot\text{lb}$

Between $y=3$ and $y=10$, gravity acts on the ball with constant -2 lb of force, and the ball moves $+7$ ft. So the work done is $(-2 \text{ lb})(7 \text{ ft}) = -14 \text{ ft} \cdot \text{lb}$.

Between $y=10$ and $y=0$, gravity acts on the ball with constant -2 lb of force, and the ball moves -10 ft. So the work done is $(-2 \text{ lb})(-10 \text{ ft}) =$

$20 \text{ ft} \cdot \text{lb}$

How much work is needed to lift a 3 kg weight 2 m off the ground?

We need to use a force that is equal to and opposite that of gravity.

Force of gravity $= (3\text{ kg})(-9.8\text{ m/s}^2) = -29.4\text{ N}$

Work needed $= (29.4\text{ N})(2\text{ m}) = 58.8\text{ J}$

A 100 ft long, 200 lb rope hangs from a cliff. How much work is needed to pull all the rope up the cliff?

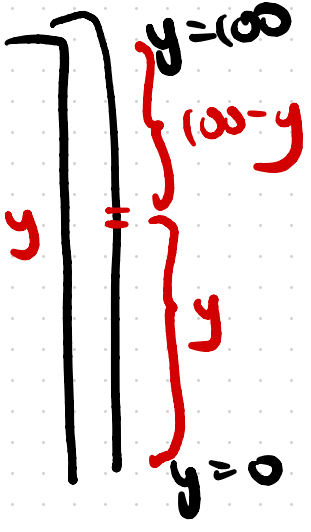
The rope is large compared to the distance moved. Different parts of the rope need to move different distances.

Treat the rope as an interval $[0, 100]$ in the y-direction. Subdivide the rope into smaller subintervals of length Δy.

The rope has weight density 200 lb per 100 ft

2 lb/ft

Each subinterval has length Δy ft, so has weight $(2 \text{ lb/ft})(\Delta y \text{ ft}) = 2\Delta y$ lb.



Let the bottom of the rope be at $y=0$, and let "up" be the positive direction.

The piece at height y needs to move a distance of $100-y$.

The work done on a piece at height y is

$$(2\,\Delta y\ \text{lb})(100-y\ \text{ft})$$

$$2(100-y)\,\Delta y \quad \text{ft}\cdot\text{lb}$$

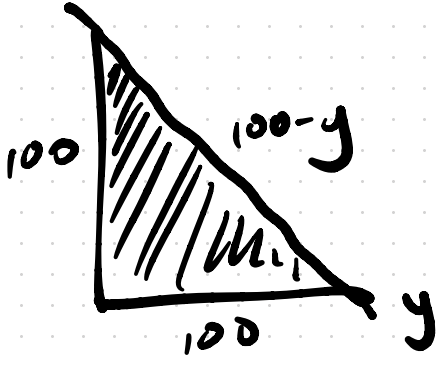
Total work done on the rope is

$$\approx \sum_{i=1}^{n} 2(100-y_i^*)\,\Delta y$$

where y_i^* is a sample point chosen from ith interval

Letting $n \to \infty$, we get the exact work done:

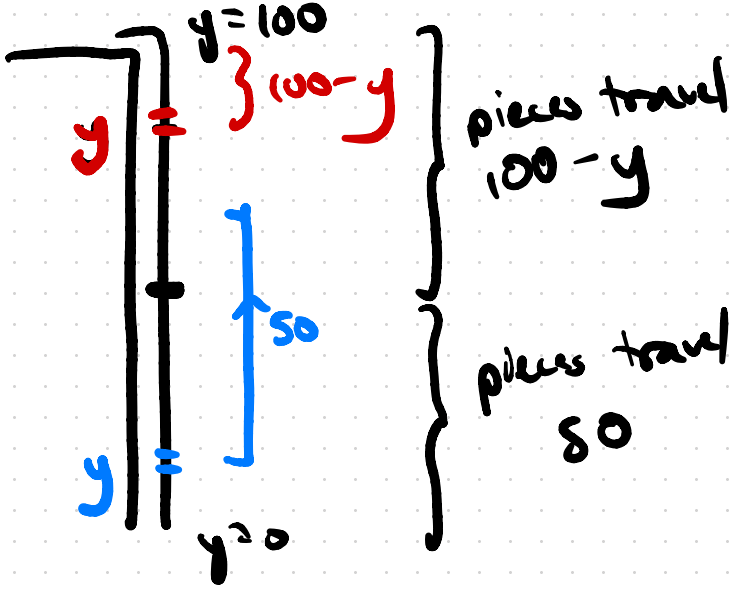
$$\int_0^{100} 2(100-y)\,dy = 2\int_0^{100} (100-y)\,dy$$



$$= 2 \cdot \frac{1}{2} \cdot 100^2$$

$$= 100^2 \text{ ft-lb}$$

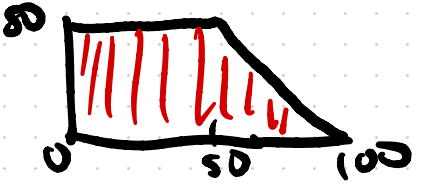
A 100 ft long, 200 lb rope hangs from a cliff. How much work is needed to pull the rope halfway up the cliff?



$$\int_0^{50} 50\,(2\,dy) + \int_{50}^{100} (100-y)(2\,dy)$$

displacement, weight

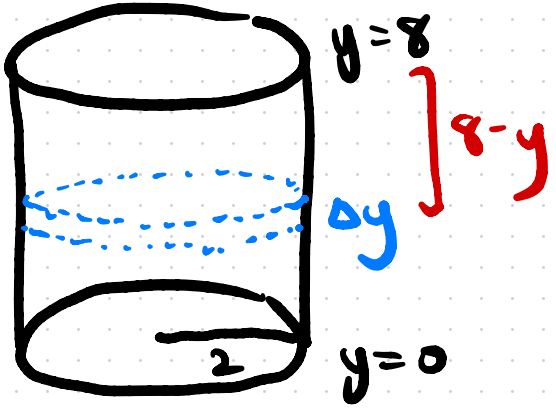
$$2\left(\int_0^{50} 50\,dy + \int_{50}^{100} (100-y)\,dy \right)$$



$2 \cdot \frac{1}{2} \cdot 7500 \text{ ft} \cdot \text{lb}$

$7500 \text{ ft} \cdot \text{lb}$

A cylindrical tank with radius 2 m and height 8 m is filled with water, which has density 1000 kg/m^3. How much work is needed to pump all the water out of the top of the tank?



Subdivide the cylinder into disks of width Δy.

Volume of disk $= \pi r^2 \Delta y$

$= 4\pi \Delta y$

Mass of disk: $(4\pi\Delta y\ m^3)(1000\ kg/m^3)$

Weight of disk: $(4\pi\Delta y\ m^3)(1000\ kg/m^3)(9.8\ m/s^2)$

$$\approx 123150\Delta y\ N$$

Displacement of a disk at height y is $8-y$ m.

Work done on disk: $(123150\Delta y)(8-y)$ J

Total work

$$\int_0^8 (8-y)(123150\,dy)$$